

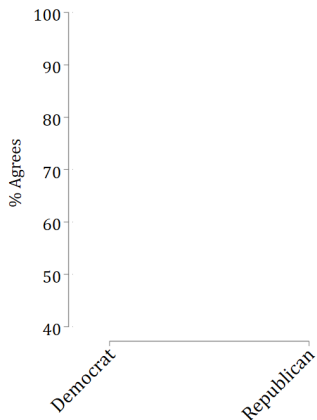
Polarized Technologies

Gaia Dossi
EIEF

Marta Morando
LSE

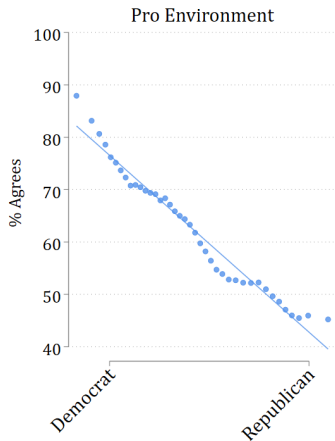
American Society is Becoming Increasingly Polarized

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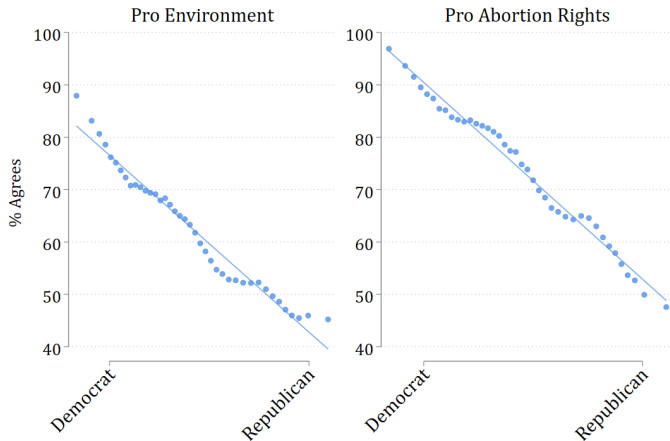
Notes. $\approx$ 500,000 US citizens from CCES (2006-2023). All plots control for age, sex, county FE, year FE, employment status FE, race FE, education FE, and income bracket FE.

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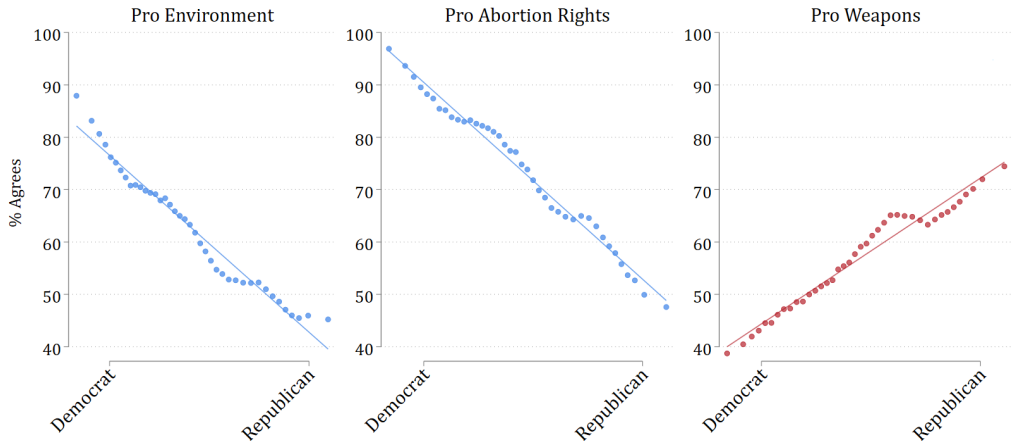
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- ▶ A large literature studies the consequences of polarization on political behavior
- ▶ Yet, implications for economic activity are less well understood

American Society is Becoming Increasingly Polarized

- ▶ A large literature studies the consequences of polarization on political behavior
- ▶ Yet, implications for economic activity are less well understood
- ▶ This paper: Impact of societal views on the direction of technological change
 1. Do views on which issues are most important shape technological change?
 2. How? Focus on how inventors' preferences or beliefs relate to direction of innovation

This Paper

1. Inventors patent technologies **aligned** with the **views** of their political party

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3. Alignment has increased **over time**, alongside rising polarization
4. In their **follow-on** work, inventors tend to build on aligned technologies
5. **Implications** for the direction of technological change

Contributions to the Literature

- ▶ Party affiliation matters for household decisions and labor market outcomes
 - Consumption (e.g., Mian et al., 2023; Conway and Boxell, 2024), investment (e.g., Meeuwis et al., 2021), health and fertility (e.g., Allcott et al., 2020; Bursztyn et al., 2022; Dahl et al., 2022), productivity (e.g., Colonnelli et al., 2022; Teso et al., 2023; Engelberg et al., 2024), on-the-job decisions (e.g., Cohen and Yang, 2019; Jelveh et al., 2024)
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- ▶ Inventor demographics matter for the direction of innovation
 - Gender (Koning et al., 2021; Einio et al., 2022), socio-economics status (Einio et al., 2022), race (Dossi, 2024), family and childhood location (Bell et al., 2018), geography (Fry, 2023; Moscona and Sastry, 2022)
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 - ⇒ **This paper:** Link between **views** and the direction of innovation
- ▶ Individuals' networks shape the diffusion of innovation
 - Networks (e.g., Jaffe et al., 1993, 2000), interaction (e.g., Singh, 2005), gender homophily (e.g., Koffi, 2024; Subramani and Saksena, 2024)
 - ⇒ **This paper:** Link inventors' views to the use of new technologies in **future** innovation

Outline

1. Data
2. Conceptual Framework
3. Main Results
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A New Dataset of Inventors Matched with Party Affiliation

- ▶ To measure inventors' views at large scale, we link inventor records to voter registers

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Last Name	First Name	Middle Name	City	ZipCode	Birth Date	Gender	Party	Registration Date
JOHN	SMITH	P	OAKLAND	941624	19250630	M	DEM	20071016

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- ▶ Final dataset: 100k unique inventors between 2001 and 2023
- ▶ Approximately even split between Democrats, Republicans, and unaffiliated

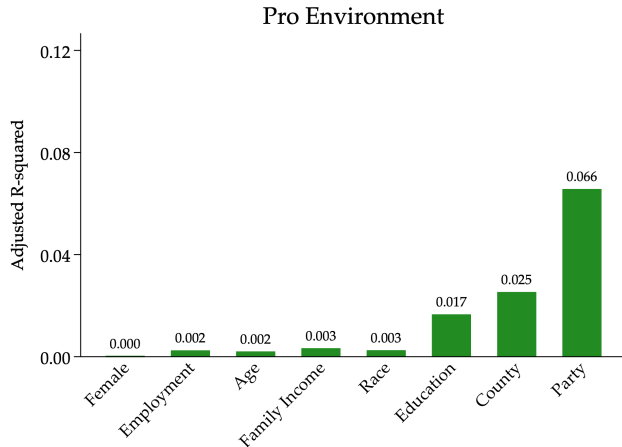
▶ Balance Table

▶ Campaign Contributions

Linking Technologies to Views of Political Parties

- ▶ Focus on **politically polarized** issues that can be mapped to technologies ▶ Topics
- ▶ Dictionary approach on patents' abstracts to define three binary variables:
 - Green (e.g., “global warming,” “anti-pollution”)
 - Female health (e.g., “endometriosis,” “ovaries”)
 - Weapons (e.g., “handgun,” “ammunition”)
- ▶ Robustness with non-dictionary measures (CPC and LLM-based)

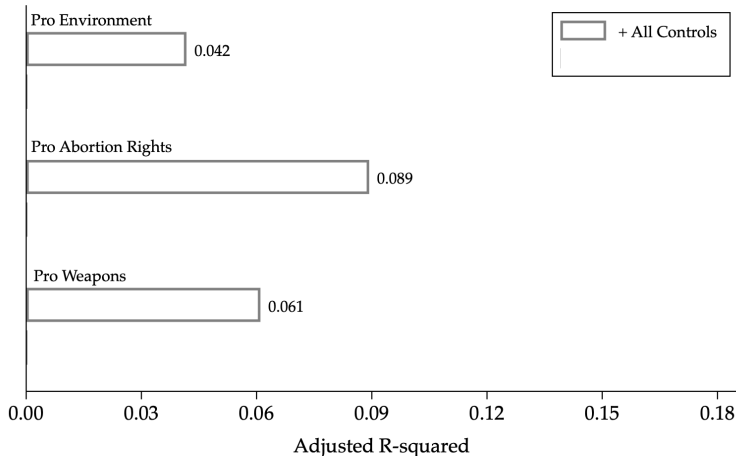
Why Focus on Political Party as a Proxy for Views?



- Support for urgent action against climate change (source: CCES, 2006-2023).

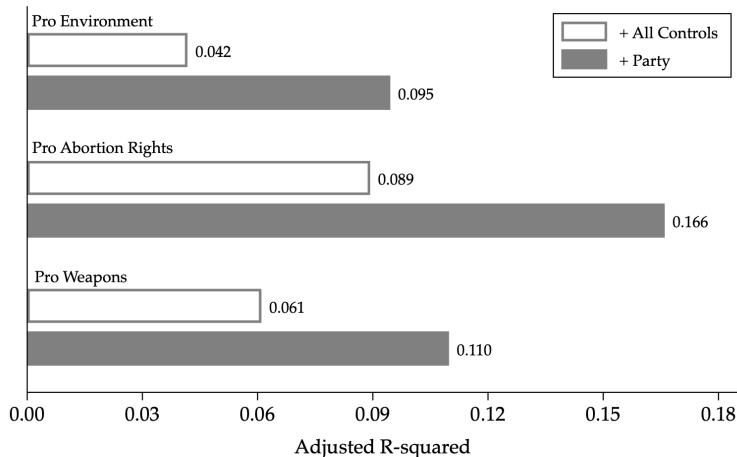
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Share Explained by Party vs All Other Demographics



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Conceptual Framework: Inventors' Technology Choice

- Inventors i choose a technology j . Each i belongs to political group $g \in \{D, R, O\}$

$$U_{ij} = w_j + \rho \cdot a_{g(i)j} + \varepsilon_{ij} \qquad j^*(i) = \arg \max_{j \in J} U_{ij}$$

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- ▶ a_{gj} = alignment of group g with technology j ; $\rho \geq 0$: salience of political alignment
- ▶ $\rho \cdot a_{g(i)j}$ is a reduced-form **perceived value** of technology j , capturing:
 1. Direct **non-monetary** utility from aligned work (e.g., Stern, 2004; Cassar and Meier, 2018)
 2. Different **beliefs** about future (financial) returns (e.g., Alesina et al., 2020)

Conceptual Framework: Inventors' Technology Choice

- ▶ Within a field, technologies offer similar **market returns** and need similar skills
- ▶ Differences in inventors' choice reflect **supply-side alignment**, not ability or returns

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- ▶ Within a field, technologies offer similar **market returns** and need similar skills
- ▶ Differences in inventors' choice reflect **supply-side alignment**, not ability or returns
- ▶ **Prediction 1.** If technology j is aligned with D's views ($a_{Dj} > a_{Rj}$):

$$\Pr(j^*(i) = j \mid g(i) = D) > \Pr(j^*(i) = j \mid g(i) = R)$$

- ▶ **Prediction 2.** As polarization rises (wider gaps $a_{Djt} - a_{Rjt}$ or higher ρ_t), gap $\uparrow$:

$$\frac{\partial}{\partial t} \left[\Pr(j^* = j \mid D) - \Pr(j^* = j \mid R) \right] > 0$$

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Empirical Specification: Inventor's Technology Choice

Define $y = 1$ if inventor i ever patents in technology j , $= 0$ otherwise:

$$y_i = \beta_1 \text{Democrat}_i + \beta_2 \text{Other}_i + \beta_3 \text{Female}_i + \gamma_{t(i)} + \delta_{c(i)} + \zeta_{s(i)} + \mu_{a(i)} + \epsilon_i$$

- ▶ t : year, c : county, s : CPC section, a : birth year
- ▶ Other_i : small parties (e.g., Independent) and unaffiliated inventors
- ▶ $\hat{\beta}_1 = \Delta$ propensity of Democrat and a Republican to ever patent technology j
- ▶ Standard errors clustered at county level

Inventors Patent Technologies Aligned With Views of their Political Party

	Green Technologies	
	(1)	(2)
Democrat	0.0024*** (0.0009)	0.0034*** (0.0009)
N. of Inventors	95,595	95,315
Scaled Difference	21.61%	30.95%
Patent Year FE	✓	✓
County FE	✓	✓
Section FE	×	✓
Birth Year FE	×	✓

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Inventors Patent Technologies Aligned With Views of their Political Party

	Green Technologies		Female-Health Technologies	
	(1)	(2)	(3)	(4)
Democrat	0.0024*** (0.0009)	0.0034*** (0.0009)	0.0037*** (0.0008)	0.0019*** (0.0007)
N. of Inventors	95,595	95,315	95,595	95,315
Scaled Difference	21.61%	30.95%	68.12%	34.70%
Patent Year FE	✓	✓	✓	✓
County FE	✓	✓	✓	✓
Section FE	×	✓	×	✓
Birth Year FE	×	✓	×	✓

Inventors Patent Technologies Aligned With Views of their Political Party

	Green Technologies		Female-Health Technologies		Weapon-related Technologies	
	(1)	(2)	(3)	(4)	(5)	(6)
Democrat	0.0024*** (0.0009)	0.0034*** (0.0009)	0.0037*** (0.0008)	0.0019*** (0.0007)	-0.0099*** (0.0014)	-0.0067*** (0.0010)
N. of Inventors	95,595	95,315	95,595	95,315	95,595	95,315
Scaled Difference	21.61%	30.95%	68.12%	34.70%	-57.74%	-39.43%
Patent Year FE	✓	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓	✓
Section FE	×	✓	×	✓	×	✓
Birth Year FE	×	✓	×	✓	×	✓

Inventors Patent Technologies Aligned With Views of their Political Party

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County FE	✓	✓	✓	✓	✓	✓
Section FE	×	✓	×	✓	×	✓
Birth Year FE	×	✓	×	✓	×	✓

- ▶ 10% gap in views is associated with a 10% gap in propensity to patent related technologies

Comparison with Match by Gender

	Green Technologies		Female-Health Technologies		Weapon-related Technologies	
	(1)	(2)	(3)	(4)	(5)	(6)
Democrat	0.0024*** (0.0009)	0.0034*** (0.0009)	0.0037*** (0.0008)	0.0019*** (0.0007)	-0.0099*** (0.0014)	-0.0067*** (0.0010)
Female		0.0013 (0.0010)		0.0068*** (0.0011)		-0.0027*** (0.0007)
N. of Inventors	95,595	95,315	95,595	95,315	95,595	95,315
Scaled Difference	21.61%	30.95%	68.12%	34.70%	-57.74%	-39.43%
Patent Year FE	✓	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓	✓
Section FE	×	✓	×	✓	×	✓
Birth Year FE	×	✓	×	✓	×	✓

► Female health: Size of match by political party is 1/3 of match by gender

Robustness Checks: Alternative Specifications

▶ Additional Fixed Effects [▶ Figure](#)

- Zip code of residence
- County-by-year
- CPC class

▶ Alternative Dependent Variables [▶ Figure](#)

- LHS defined as $= 1$ if inventor i ever patented in technology j as first author
- LHS defined as % of patents in technology j by inventor i
- Using a Poisson count model on total n. of patents in technology j by inventor i

▶ Alternative Units of Observation [▶ Table](#)

- Estimating a regression at the patent level (instead of inventor level)

Robustness Checks: Alternative Samples

- ▶ Is the match driven by inventors adopting the views of the firm? [▶ Figure](#)
 - Similar results for sample who **registered young** (before entering labor market)
- ▶ Is the match driven by patent examiners selectively granting patents? [▶ Figure](#)
 - Similar results in sample of patent **applications**
- ▶ Is the match limited to four US states? [▶ Figure](#)
 - Similar results across **all US states** using Campaign Contributions data

Outline

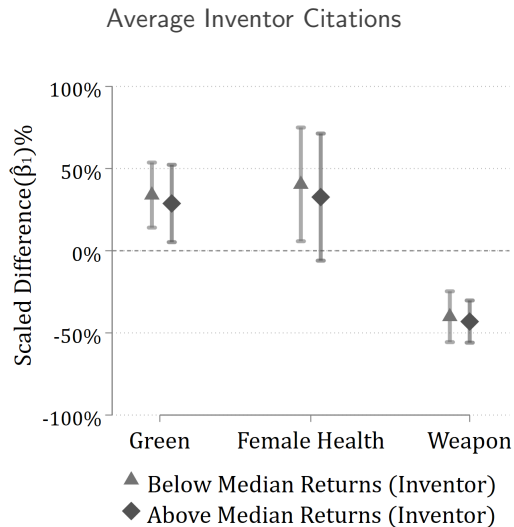
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The Role of Inventor Skills

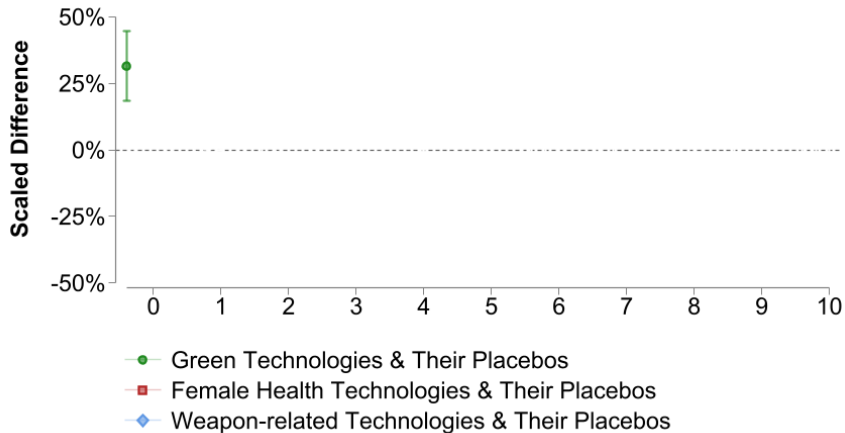
Match with Polarized Technologies Persists Across Inventor Productivities



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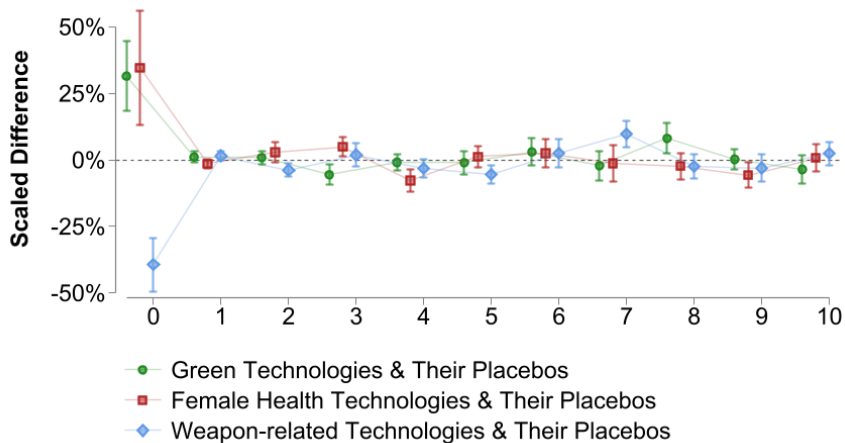
Technology-Specific Skills Do Not Explain the Match



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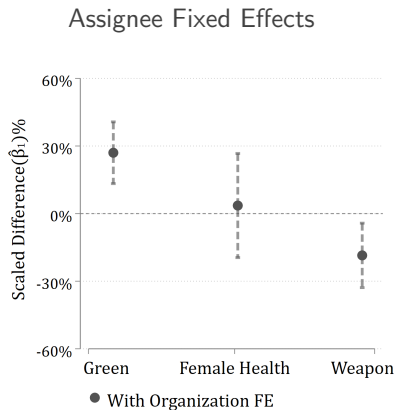


Technology-Specific Skills Do Not Explain the Match



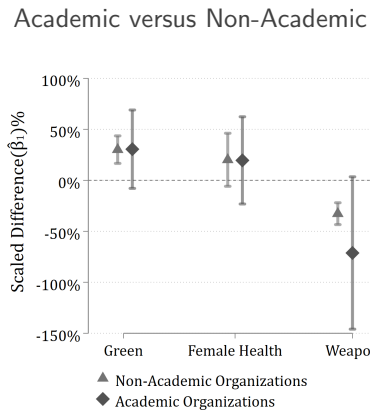
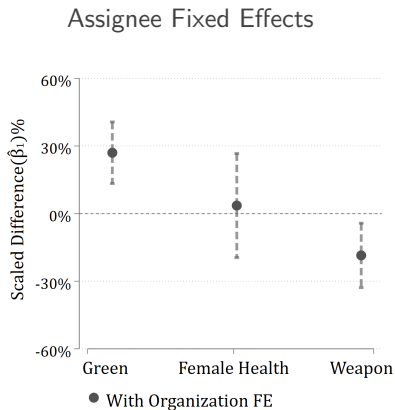
The Role of Organizations

Match with Polarized Technologies Persists Within Assignees [► Other Results](#)



- Match holds within assignees (caveat: sample only includes large assignees)

Match with Polarized Technologies Persists Within Assignees ► Other Results

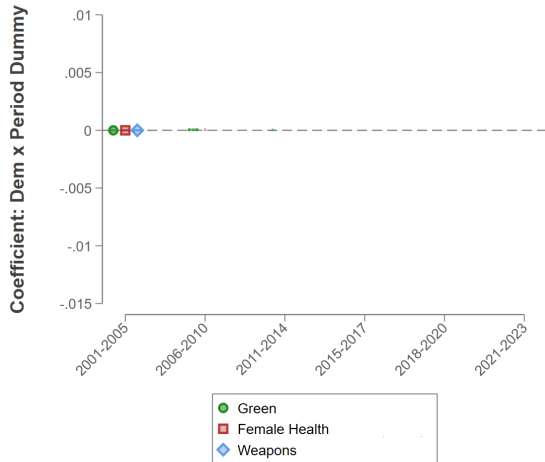


- Match holds within assignees (caveat: sample only includes large assignees)
- Also holds in universities: inventors have more freedom to choose direction of research

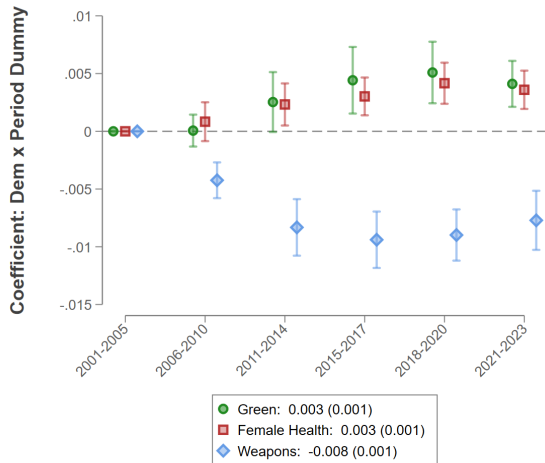
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The Match with Polarized Technologies Has Increased Over Time



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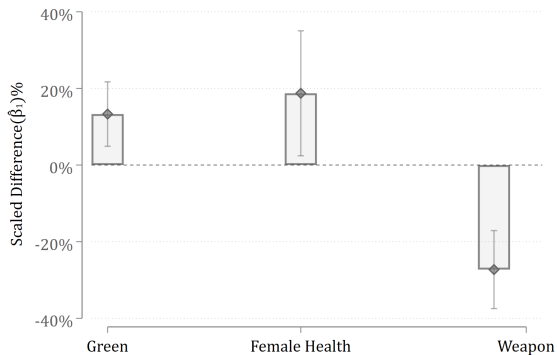
Polarized Technologies and Follow-On Innovation

The Use of Polarized Technologies in Follow-On Innovation

$$y_i = \beta_1 \text{Democrat}_i + \beta_2 \text{Other}_i + \beta_3 \text{Female}_i + \gamma_{t(i)} + \delta_{c(i)} + \zeta_{s(i)} + \mu_{a(i)} + \epsilon_i$$

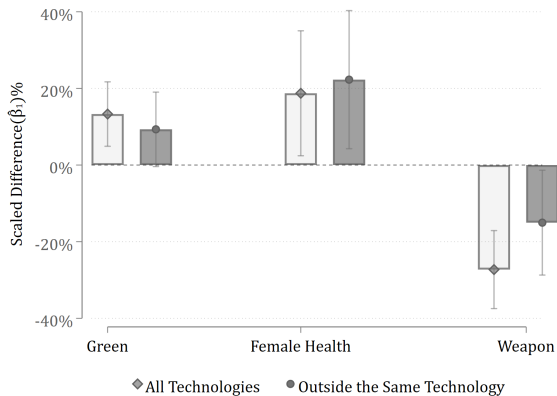
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Policy Implications

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1. Polarization amplifies sorting: if group sized differ, aggregate direction may change
 - Costlier to redirect inventive effort across party lines (alignment acts as a friction)

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1. Polarization amplifies sorting: if group sized differ, aggregate direction may change
 - Costlier to redirect inventive effort across party lines (alignment acts as a friction)
2. Shifting societal views can lead to endogenous technological change
 - Through **supply** of new technologies, not only demand [Besley Persson 2023]

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Conclusion

- ▶ Societal views and their polarization can shape **direction** of technological change
- ▶ They operate at least in part through inventors' **technology choices**
- ▶ Technological progress is a product of the **ideological environment** [Mokyr, 2016]

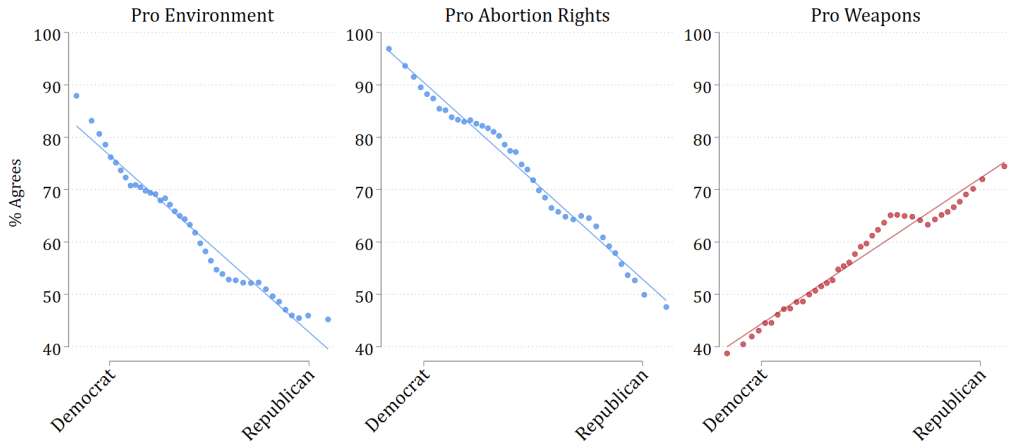
Thank You!

Additional Materials

Topics Covered in the CCES – Political Attitudes Module [◀ Back](#)

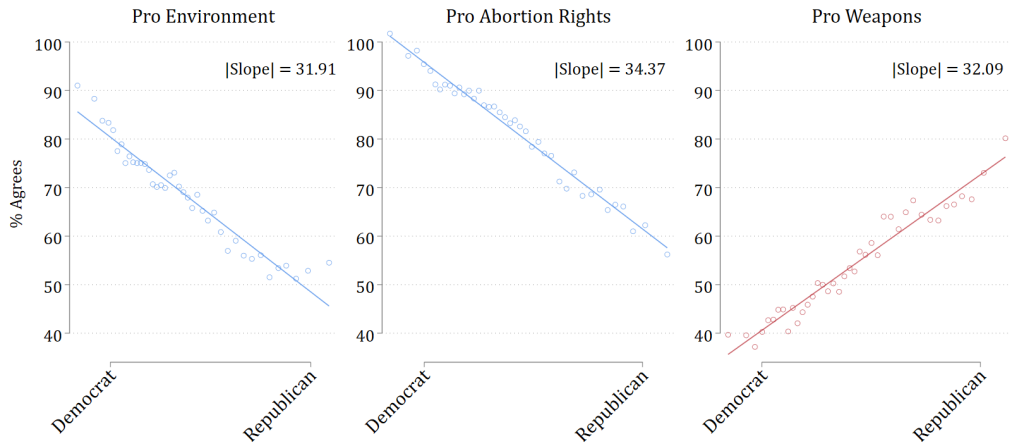
- ▶ Abortion
- ▶ Environment
- ▶ Guns
- ▶ Health care
- ▶ Immigration
- ▶ Military
- ▶ Government spending
- ▶ Trade
- ▶ Gay marriage
- ▶ Affirmative action
- ▶ Income vs sales tax

Polarizing Issues: All Individuals [◀ Back](#)



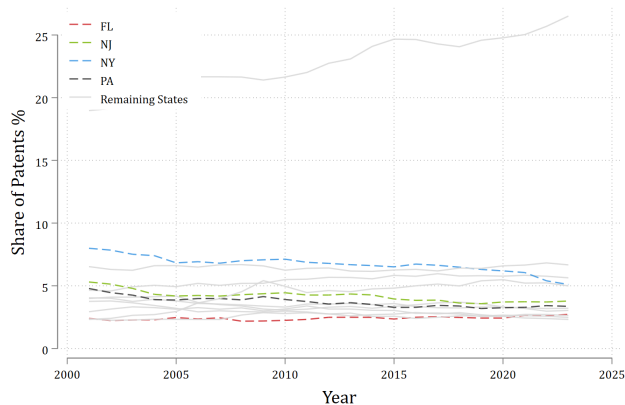
Notes. $\approx$ 500,000 US citizens from CCES (2006-2023). All plots control for age, sex, county FE, year FE, employment status FE, race FE, education FE, and income bracket FE.

Polarizing Issues: Individuals With Characteristics of Inventors [◀ Back](#)



Notes. $\approx$ 500,000 US citizens from CCES (2006-2023). All plots control for age, sex, county FE, year FE, employment status FE, race FE, education FE, and income bracket FE. Sample: individuals with more than 150k in revenues per year and college graduates.

Importance of FL and NY for Total US Innovation [◀ Back](#)



Notes. The figure plots the evolution of the yearly share of patents (by residence of inventors) for the top 10 US states in terms of innovation.

Merge Between Patent and Voters Data [◀ Back](#)

- ▶ NY 2020 (N. 19mln) + FL 2017 & 2022 (N. 16mln) + PA 2020 (N. 13mln) + NJ 2022 (N. 9 mln) voter data
- ▶ Drop those younger than 16 and older than 100 at registration + pre-clean strings in same way
- ▶ Conservative match algorithm (by state):
 1. Exact match last name, first name, and city of residence
 2. Middle initial matches exactly or missing in one of the two
 3. Remove those younger than 22 at the first or last patenting year
 4. Remove those older than 89 at the first or last patenting year
 5. Among duplicates:
 - i. keep those with the same middle initials
 - ii. keep those with same party (as Teso et al. (2023))
 - iii. keep matches randomly
 6. Results unchanged if we keep only exact matches
- ▶ $\approx$ 53% of patents matched
- ▶ $\approx$ 8% of all US patents since 2001

Balance Table Matched-Unmatched Inventors [◀ Back](#)

H_0 = difference in characteristic X is larger than $10\% \times SD(X)$

	Matched		Unmatched		Matched-Unmatched	
	Mean (1)	Standard Deviation (2)	Mean (3)	Standard Deviation (4)	Standardized Difference (5)	P-value Equivalence Test (6)
Female	0.134	0.341	0.151	0.358	-0.047	0.000
N. Consonants First Name	3.682	1.141	3.609	1.254	0.060	0.000
N. Consonants Middle Name	0.826	1.284	0.726	1.282	0.078	0.000
N. Consonants Last Name	4.138	1.400	4.032	1.575	0.070	0.000
Length First Name	5.842	1.516	5.820	1.754	0.013	0.000
Length Middle Name	1.201	1.976	1.078	1.998	0.062	0.000
Length Last Name	6.489	1.970	6.429	2.334	0.027	0.000
City-level Income (Log)	12.680	1.397	12.580	1.364	0.073	0.000
City-level Population (Log)	11.030	1.819	10.990	1.711	0.023	0.000
A Section	0.285	0.435	0.291	0.443	-0.012	0.000
B Section	0.197	0.368	0.172	0.358	0.068	0.000
C Section	0.162	0.349	0.192	0.383	-0.083	0.000
D Section	0.010	0.087	0.010	0.091	-0.001	0.000
E Section	0.043	0.191	0.034	0.174	0.047	0.000
F Section	0.100	0.281	0.081	0.260	0.072	0.000
G Section	0.366	0.452	0.358	0.459	0.018	0.000
H Section	0.235	0.395	0.260	0.420	-0.061	0.000
Y Section	0.151	0.316	0.151	0.329	0.001	0.000
Grant Year	2014	6.358	2013	6.822	0.149	1.000
N. Patent Citations	14.960	39.230	18.490	50.520	-0.075	0.000

Difference between Republican & Democrat Inventors [◀ Back](#)

	Democrat - Republican (Unconditional)			Democrat - Republican (Conditional)		
	Coefficient (1)	Standard Error (2)	P-value (3)	Coefficient (4)	Standard Error (5)	P-value (6)
Female Dummy	0.095	0.003	0.000	0.081	0.003	0.000
Birth Year	2.978	0.106	0.000	1.400	0.172	0.000
Income	14591.510	350.618	0.000	-1803.527	1144.451	0.117
A Section	0.047	0.004	0.000	0.023	0.006	0.000
B Section	-0.077	0.003	0.000	-0.042	0.004	0.000
C Section	0.080	0.003	0.000	0.061	0.007	0.000
D Section	-0.001	0.001	0.317	-0.000	0.001	1.000
E Section	-0.038	0.002	0.000	-0.025	0.002	0.000
F Section	-0.068	0.003	0.000	-0.042	0.004	0.000
G Section	0.100	0.004	0.000	0.047	0.006	0.000
H Section	0.025	0.003	0.000	0.015	0.006	0.013
Y Section	-0.029	0.003	0.000	-0.013	0.004	0.001

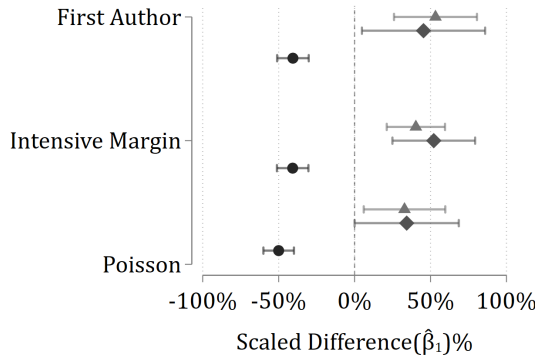
Inventors Patent Technologies Aligned With Views of Their Party [◀ Back](#)

	Green Technologies			Female-Health Technologies			Weapon-related Technologies		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Democrat $\hat{\beta}_1$	0.0024*** (0.0009)	0.0036*** (0.0008)	0.0034*** (0.0009)	0.0037*** (0.0008)	0.0023*** (0.0007)	0.0019*** (0.0007)	-0.0099*** (0.0014)	-0.0067*** (0.0010)	-0.0067*** (0.0010)
Other $\hat{\beta}_2$	0.0016* (0.0009)	0.0019** (0.0009)	0.0019** (0.0009)	0.0020** (0.0009)	0.0012 (0.0008)	0.0014* (0.0008)	-0.0057*** (0.0010)	-0.0038*** (0.0009)	-0.0040*** (0.0009)
Female $\hat{\beta}_3$			0.0013 (0.0010)			0.0068*** (0.0011)			-0.0027*** (0.0007)
$\hat{\beta}_2 - \hat{\beta}_1$	-0.0008	-0.0016	-0.0015	-0.0017	-0.0010	-0.0005	0.0041	0.0028	0.0028
P-value $\hat{\beta}_2 - \hat{\beta}_1$	[0.3527]	[0.0560]	[0.0881]	[0.0170]	[0.1394]	[0.5072]	[0.0000]	[0.0007]	[0.0006]
N. of Inventors	95,595	95,595	95,315	95,595	95,595	95,315	95,595	95,595	95,315
% of Dem.	35.78	35.78	35.79	35.78	35.78	35.79	35.78	35.78	35.79
$\mathbb{E}(LHS)$ for Rep.	0.011	0.011	0.011	0.005	0.005	0.005	0.017	0.017	0.017
Scaled Difference (%)	21.61	32.49	30.95	68.12	41.63	34.70	-57.74	-39.05	-39.43
Patent Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Section FE	×	✓	✓	×	✓	✓	×	✓	✓
Birth Year FE	×	×	✓	×	×	✓	×	×	✓

Comparison With Match by Gender: Split by Male and Female [◀ Back](#)

	Green Technologies			Female-Health Technologies			Weapon-related Technologies		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Male Sample									
Democrat $\hat{\beta}_1$	0.0027*** (0.0010)	0.0037*** (0.0010)	0.0036*** (0.0010)	0.0026*** (0.0008)	0.0015** (0.0007)	0.0015** (0.0007)	-0.0101*** (0.0014)	-0.0070*** (0.0011)	-0.0072*** (0.0011)
N. of Inventors	82,547	82,547	82,547	82,547	82,547	82,547	82,547	82,547	82,547
% of Dem.	33.77	33.77	33.77	33.77	33.77	33.77	33.77	33.77	33.77
$\mathbb{E}(LHS)$ for Rep.	0.011	0.011	0.011	0.005	0.005	0.005	0.018	0.018	0.018
Scaled Difference %	23.75	32.31	31.69	54.77	31.89	32.25	-55.38	-38.45	-39.77
Panel B: Female Sample									
Democrat $\hat{\beta}_1$	0.0018 (0.0018)	0.0015 (0.0017)	0.0013 (0.0017)	0.0053** (0.0025)	0.0048* (0.0025)	0.0047* (0.0025)	-0.0038** (0.0015)	-0.0032** (0.0014)	-0.0033** (0.0014)
N. of Inventors	12,738	12,738	12,736	12,738	12,738	12,736	12,738	12,738	12,736
% of Dem.	48.81	48.81	48.8	48.81	48.81	48.8	48.81	48.81	48.8
$\mathbb{E}(LHS)$ for Rep.	0.006	0.006	0.006	0.013	0.013	0.013	0.006	0.006	0.006
Scaled Difference (%)	31.25	26.30	22.16	40.09	36.29	35.87	-62.21	-52.21	-54.07

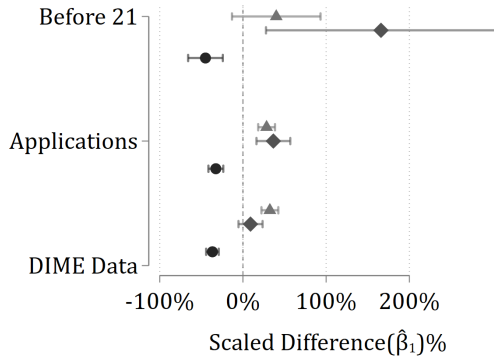
Robustness Checks: Alternative Dependent Variables [◀ Back](#)



Robustness Checks: Alternative Specification [◀ Back](#)

	Solo-Authored			Teams			Homogeneous		
	Green (1)	Female Health (2)	Weapon-related (3)	Green (4)	Female Health (5)	Weapon-related (6)	Green (7)	Female Health (8)	Weapon-related (9)
Democrat $\hat{\beta}_1$	0.0031** (0.0016)	0.0012 (0.0009)	-0.0074*** (0.0019)	0.0056*** (0.0013)	0.0017 (0.0012)	-0.0038*** (0.0012)	0.0051*** (0.0012)	0.0013* (0.0008)	-0.0029*** (0.0011)
N. of Patents	53,189	53,189	53,189	122,026	122,026	122,026	122,026	122,026	122,026
% of Dem.	31.26	31.26	31.26	37.11	37.11	37.11	23.29	23.29	23.29
$\mathbb{E}(LHS)$ for Rep.	0.005	0.003	0.019	0.006	0.005	0.004	0.004	0.002	0.007
Scaled Difference (%)	60.44	42.44	-40.03	88.23	37.89	-88.85	134.73	53.40	-40.54

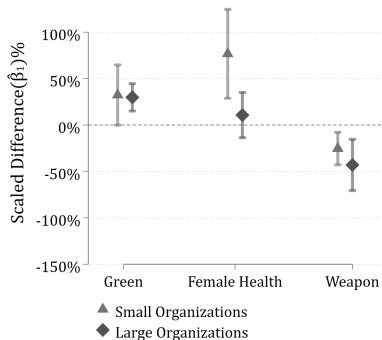
Robustness Checks: Alternative Samples [◀ Back](#)



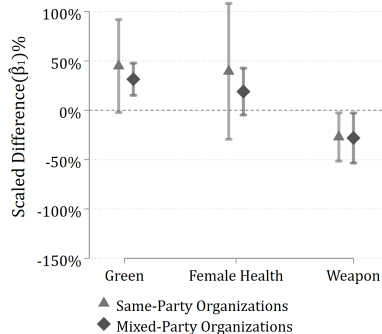
Match with Polarized Technologies Persists Across Assignee Characteristics

[◀ Back](#)

Small versus Large



Same-Party versus Mixed-Party



- ▶ Match holds in small assignees: unlikely to be due to networks in hiring (Colonnelli et al., 2022)
- ▶ Also holds in politically-homogeneous assignees: unlikely to be driven by homophily in hiring

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